The partially symmetric rank of products of binary monomials

TBD

Abstract

The known upper bound for products of generalized W tensors suggests a matching lower bound on every decomposition with the symmetric blocks fixed. We obtain it by normalizing coordinates, matching coefficients as moment conditions, and constructing a weighted bilinear Gram matrix of explicit test functions. Invertibility of this matrix forces the number of summands to be at least the number of test functions. Exact Laurent coefficients prove nondegeneracy in one finite choice of parameters, including for decompositions with repeated nodes and zero coordinates. More generally, for every $k \geq 1$ and integers $a_i \geq b_i \geq 1$, we prove that the complex partially symmetric rank is

$$\mathrm{R}\left(\prod_{i=1}^{k} x_i^{a_i} y_i^{b_i}\right) = \prod_{i=1}^{k}(b_i+1) + \sum_{i=1}^{k}(a_i - b_i)\prod_{j\neq i}(b_j+1).$$

A finite-character construction supplies the matching upper bound. For generalized W tensors of orders $d_i \geq 3$, the formula becomes $2^{k-1}(\sum_i d_i - 2k + 2)$, attaining the upper bound of Canino–Casarotti–Santarsiero. Three cubic W tensors have rank 20, and three copies of x^3y^2 have rank 54.

1 Rank, the main result, and its context

For $k \geq 2$ and $d_i \geq 3$, Canino–Casarotti–Santarsiero's construction gives the upper bound $M = 2^{k-1}(\sum_i d_i - 2k + 2)$ for the fixed-block partially symmetric rank of $\prod_i x_i^{d_i-1} y_i$ [3, Theorem 1.1]. To prove the matching lower bound, discard zero summands from any proposed decomposition and let r count the remaining terms. An invertible change of coordinates makes its factors affine, with their scales absorbed into nonzero weights. Matching every coefficient gives product-moment identities. By linearity these determine weighted evaluation on the polynomial space with coordinate degrees at most d_i.

The proof constructs M test functions in batches $\mathcal{B}_0, \mathcal{B}_1, \ldots, \mathcal{B}_k$, whose evaluations form an $r \times M$ matrix E. With an auxiliary diagonal weight matrix W, their Gram matrix is $G = E^{\mathsf{T}} W E$. Once G is invertible, E has independent columns and $r \geq M$. The main step is to make every diagonal Gram block nondegenerate at one finite parameter vector; orthogonality alone is insufficient. The product-moment proof below establishes this step in the broader binary-monomial setting and retains its full parameter range.

Let $k \geq 1$, let $d_1, \ldots, d_k$ be positive integers, and identify

$$\bigotimes_{i=1}^{k} \mathrm{Sym}^{d_i}(\mathbb{C}^2)$$

with the polynomials in disjoint pairs (x_i, y_i) that have degree d_i in the ith pair. For a nonzero such polynomial F, its fixed-block partially symmetric rank $\mathrm{R}_{\mathbf{d}}(F)$ is the least n such that

$$F = \sum_{\ell=1}^{n} c_\ell \prod_{i=1}^{k} (u_{i,\ell} x_i + v_{i,\ell} y_i)^{d_i}, \qquad c_\ell, u_{i,\ell}, v_{i,\ell} \in \mathbb{C}. \tag{1.1}$$

The degrees and the k blocks remain fixed. Coefficients may be absorbed into a factor by extracting a complex d_ith root, so this definition agrees with the definition using sums of partially symmetric pure tensors without displayed scalar coefficients. Multiplication of F by a nonzero scalar also leaves its rank unchanged. We suppress the subscript $\mathbf{d}$ when the degrees are clear.

Theorem 1.1. *For every integer $k \geq 1$ and integers $a_i \geq b_i \geq 1$, set $d_i = a_i + b_i$. Then, over $\mathbb{C}$,*

$$\mathrm{R}_{\mathbf{d}}\left(\prod_{i=1}^{k} x_i^{a_i} y_i^{b_i}\right) = N(\mathbf{a}, \mathbf{b}) := \prod_{i=1}^{k} (b_i + 1) + \sum_{i=1}^{k} (a_i - b_i) \prod_{j \neq i} (b_j + 1). \tag{1.2}$$

Empty products have value 1. The lower bound holds for every finite decomposition of the form (1.1), without restrictions on the complex coefficients or support. The upper bound is realized by the explicit finite-character identity (4.5).

The theorem concerns exact complex rank with fixed symmetric blocks. No assertion about real rank, border rank, cactus rank, unrestricted nonsymmetric rank, or rank after grouping modes from different blocks is included. The lower argument uses only finite decompositions and does not apply to limits of decompositions.

For $d \geq 3$, the generalized W tensor is

$$W_d = \sum_{j=1}^{d} e_1^{\otimes(j-1)} \otimes e_2 \otimes e_1^{\otimes(d-j)} \in \mathrm{Sym}^d(\mathbb{C}^2).$$

Under the usual polynomial identification this is a nonzero scalar multiple of $x^{d-1}y$.

Corollary 1.2. *For every $k \geq 2$ and integers $d_i \geq 3$, the partially symmetric rank of the product of generalized W tensors is*

$$\mathrm{R}_{\mathbf{d}}(W_{d_1} \otimes \cdots \otimes W_{d_k}) = 2^{k-1} \left(\sum_{i=1}^{k} d_i - 2k + 2 \right). \tag{1.3}$$

In particular $\mathrm{R}_{(3,3,3)}(W_3^{\otimes 3}) = 20$ *and* $\mathrm{R}_{(3,\ldots,3)}(W_3^{\otimes k}) = 2^{k-1}(k+2)$.

Proof. Take $a_i = d_i - 1$ and $b_i = 1$ in Theorem 1.1, and use invariance of rank under nonzero scalar multiplication. □

1.1 Prior work

Oneto and Ventura ask for the partially symmetric rank of $W_{d_1} \otimes \cdots \otimes W_{d_k}$ with $d_i \geq 3$ in [1, Question 12]. Here W_d corresponds, up to a nonzero scalar, to $x^{d-1}y$, and each copy remains a separate symmetric block. Ballico, Bernardi, Christandl and Gesmundo proved

$$\mathrm{R}(W_3^{\otimes k}) \leq (k+2)2^{k-1} \quad \text{and} \quad \mathrm{R}(W_{d_1} \otimes \cdots \otimes W_{d_k}) \leq 2^{k-1} \sum_{i=1}^{k} d_i,$$

the latter for $k \geq 2$ and $d_i \geq 3$ [2, Theorems 3.3 and 3.6]. Their Proposition 4.3 gives the lower bound $\sum_i d_i - k + 1$ already for ordinary tensor rank, hence also for partially symmetric rank. Canino, Casarotti and Santarsiero subsequently obtained

$$\mathrm{R}(W_{d_1} \otimes \cdots \otimes W_{d_k}) \leq 2^{k-1} \left(\sum_{i=1}^{k} d_i - 2k + 2 \right) \qquad (k \geq 2,\ d_i \geq 3)$$

by an explicit construction [3, Theorem 1.1].

For two binary-monomial blocks, Gałązka constructed a reduced apolar set giving

$$\mathrm{R}(x_1^{a_1} y_1^{b_1} x_2^{a_2} y_2^{b_2}) \leq (a_1+1)(b_2+1) + (b_1+1)(a_2+1) - (b_1+1)(b_2+1), \qquad a_i \geq b_i \geq 1,$$

and obtained lower bounds and exact special cases [4, Theorem 1.5 and Section 7.1 of the arXiv version]. The binomial equations in that construction describe a finite set in the two-dimensional torus. The lower bound

$$\mathrm{R}(x_1^{a_1} y_1^{b_1} x_2^{a_2} y_2^{b_2}) \geq \max\{(a_1+1)(b_2+1), (b_1+1)(a_2+1)\}$$

also appears in [5, Proposition 4.3 of the arXiv version].

Behr's unpublished manuscript [6] gives the exact two-block formula

$$\mathrm{R}(x_1^{a_1} y_1^{b_1} x_2^{a_2} y_2^{b_2}) = (a_1+1)(a_2+1) - (a_1-b_1)(a_2-b_2) \qquad (a_i \geq b_i \geq 1)$$

in Corollary 5.4, including $\mathrm{R}(W_{d_1} \otimes W_{d_2}) = 2(d_1+d_2-2)$ for $d_1, d_2 \geq 2$ in Theorem 1.1. Thus the two-block specialization is already present in that work. Its section on three or more factors leaves that case open.

Related decomposition questions. SOS tensor decompositions write a real homogeneous polynomial of even degree as a sum of squares of real homogeneous polynomials of half that degree; SOS-rank counts the smallest number of those squares [7, Section 4]. Here the admissible terms are products of powers of complex linear forms in prescribed symmetric blocks. These are different decomposition problems, and their ranks are not identified. No SOS decomposition theorem is used in either bound.

1.2 The two constructions

The lower bound will be proved first for arbitrary finite product-moment formulas. The moments vanish above a prescribed degree b_i in each coordinate, while the moment at degree b_i is nonzero. There are $Q = \prod_i (b_i+1)$ basic evaluation vectors. For each coordinate an additional list contributes $(d_i - 2b_i)Q/(b_i+1)$ vectors. In the original coordinates the corresponding Gram matrices can be singular, so orthogonality alone does not justify adding dimensions.

We introduce parameters t_i and replace the diagonal pairing weights w_ℓ by $w_\ell \prod_i (1 + t_i z_{i,\ell})^{d_i}$. Suitable rational evaluation vectors remain orthogonal in distinct lists. Within each additional list, reversing block columns gives a triangular Gram matrix. Its diagonal block has an invertible leading Laurent coefficient. Each determinant is therefore a nonzero rational function, and one finite parameter vector avoids all its zeros and poles.

For the upper bound, write $q_i = b_i + 1$ and $\delta_i = a_i - b_i$. The column lattice of $A = \mathrm{diag}(q_1, \ldots, q_k) + \delta \mathbf{1}^T$ has index N and meets $\prod_i [-a_i, b_i]$ only at the origin. Summing over the characters of $\mathbb{Z}^k / A\mathbb{Z}^k$ then selects exactly the desired monomial among all coefficients of a sum of partially symmetric powers. Both bounds are proved here; the tensor-rank results cited for context are not used as mathematical premises.

2 A product-moment lower bound

The lower bound is a statement about finite product-moment formulas over $\mathbb{C}$. No positivity of the weights, distinctness of the nodes, or nonvanishing of their coordinates is required.

Theorem 2.1 (Product-moment lower bound)**.** *Let $k \geq 1$, and let $b_i \geq 1$ and $d_i \geq 2b_i$ be integers for $1 \leq i \leq k$. Suppose that*

$$m_i : \mathbb{C}[z_i]_{\leq d_i} \longrightarrow \mathbb{C}$$

is a linear functional satisfying

$$m_i(z_i^e) = 0 \quad (b_i < e \leq d_i), \qquad \beta_i := m_i(z_i^{b_i}) \neq 0. \tag{2.1}$$

Let $n \geq 1$, and let $z_\ell = (z_{1,\ell}, \ldots, z_{k,\ell}) \in \mathbb{C}^k$ and $w_\ell \in \mathbb{C} \setminus \{0\}$, for $1 \leq \ell \leq n$, satisfy

$$\sum_{\ell=1}^{n} w_\ell \prod_{i=1}^{k} z_{i,\ell}^{e_i} = \prod_{i=1}^{k} m_i(z_i^{e_i}) \qquad (0 \leq e_i \leq d_i). \tag{2.2}$$

Set

$$q_i = b_i + 1, \qquad \delta_i = d_i - 2b_i, \qquad Q = \prod_{i=1}^{k} q_i, \qquad N = Q + \sum_{i=1}^{k} \delta_i \frac{Q}{q_i}. \tag{2.3}$$

Then $n \geq N$. The nodes may repeat, and any of their coordinates may be zero.

We prove the theorem by constructing N vectors with an invertible Gram matrix. The construction uses a parameter-dependent symmetric bilinear pairing and indexed evaluation lists. We first establish nonsingularity over a rational function field, then choose one common finite parameter vector.

Introduce independent parameters $t_1, \ldots, t_k$, and write

$$K = \mathbb{C}(t_1, \ldots, t_k), \qquad h_{i,\ell} = 1 + t_i z_{i,\ell}, \qquad w_\ell(t) = w_\ell \prod_{i=1}^{k} h_{i,\ell}^{d_i}.$$

Each $h_{i,\ell}$ is a nonzero polynomial, including when $z_{i,\ell} = 0$. On K^n use the diagonal bilinear pairing

$$\langle f, g \rangle_t = \sum_{\ell=1}^{n} w_\ell(t) f(\ell) g(\ell). \tag{2.4}$$

For $0 \leq \varepsilon_i \leq b_i$, define

$$u_\varepsilon(\ell) = \prod_{i=1}^{k} \frac{z_{i,\ell}^{\varepsilon_i}}{h_{i,\ell}^{b_i}}, \qquad \mathcal{U} = (u_\varepsilon)_{0 \leq \varepsilon_i \leq b_i}. \tag{2.5}$$

For each i with $\delta_i > 0$, also define

$$v_{i,s,\varepsilon}(\ell) = \frac{z_{i,\ell}^{s}}{h_{i,\ell}^{s}} \prod_{j \neq i} \frac{z_{j,\ell}^{\varepsilon_j}}{h_{j,\ell}^{b_j}}, \quad \begin{array}{l} b_i + 1 \leq s \leq d_i - b_i, \\ 0 \leq \varepsilon_j \leq b_j \quad (j \neq i). \end{array} \tag{2.6}$$

Denote this list by $\mathcal{V}_i$; when $\delta_i = 0$, let $\mathcal{V}_i$ be empty. Thus $\mathcal{U}$ contains Q vectors and $\mathcal{V}_i$ contains $\delta_i Q / q_i$ vectors. Empty products, including empty Kronecker products of matrices, are interpreted as 1.

By linearity, (2.2) applies to any product of univariate polynomials whose degree in coordinate i is at most d_i. We use it in this form throughout the Gram calculations.

Lemma 2.2. *For $0 \le p, r \le b_i$, set*

$$G_i(t_i)_{p,r} = m_i(z_i^{p+r}(1+t_i z_i)^{\delta_i}).$$

The Gram matrix of $\mathcal{U}$ is $\bigotimes_{i=1}^{k} G_i(t_i)$, and

$$\det G_i(t_i) = (-1)^{q_i(q_i-1)/2}\beta_i^{q_i} \neq 0. \tag{2.7}$$

In particular, this Gram matrix is invertible for every finite choice of the parameters for which the vectors are defined.

Proof. The polynomial defining an entry of G_i has degree at most $2b_i + \delta_i = d_i$. If $p + r > b_i$, every monomial in it has zero m_i-value by (2.1). If $p + r = b_i$, only its lowest term survives, and the entry equals β_i. Reversing the columns therefore gives an upper triangular matrix with constant diagonal β_i, proving (2.7). Finally, (2.4) and (2.5) give

$$\langle u_\varepsilon, u_\eta\rangle_t = \sum_\ell w_\ell \prod_i z_{i,\ell}^{\varepsilon_i+\eta_i} h_{i,\ell}^{\delta_i} = \prod_i G_i(t_i)_{\varepsilon_i,\eta_i},$$

which identifies the Gram matrix as the stated Kronecker product. □

Lemma 2.3. *The lists $\mathcal{U}, \mathcal{V}_1, \dots, \mathcal{V}_k$ are mutually orthogonal for (2.4).*

Proof. In pairing $v_{i,s,\varepsilon}$ with u_η, the factor in coordinate i, after including the weight, is

$$z_i^{s+\eta_i}(1+t_i z_i)^{d_i-s-b_i}.$$

The exponent of $1 + t_i z_i$ is nonnegative. Every monomial has degree at least $s > b_i$ and at most $d_i - b_i + \eta_i \le d_i$, so its m_i-value is zero. Every other coordinate also has degree at most its prescribed d_j. Equation (2.2) proves $\langle \mathcal{V}_i, \mathcal{U}\rangle_t = 0$.

If $i \neq j$, pair a vector from $\mathcal{V}_i$ with one from $\mathcal{V}_j$. The coordinate-i factor has the same form, with η_i supplied by the second vector, and again has zero moment. The coordinate-j factor has nonnegative exponent $d_j - s' - b_j$ on $1 + t_j z_j$ and degree at most d_j. All remaining coordinates have the form used in Lemma 2.2. Thus every use of (2.2) stays inside its degree box, and $\langle \mathcal{V}_i, \mathcal{V}_j\rangle_t = 0$. □

It remains to analyze the Gram matrix within each nonempty $\mathcal{V}_i$. Put $r_i = Q/q_i$, and let $H_i(t)$ be the r_i-by-r_i matrix indexed by the tuples ε, η on coordinates other than i, with entries

$$H_i(t)_{\varepsilon,\eta} = \sum_{\ell=1}^{n} w_\ell \frac{z_{i,\ell}^{d_i+1}}{1+t_i z_{i,\ell}} \prod_{j\neq i} z_{j,\ell}^{\varepsilon_j+\eta_j}(1+t_j z_{j,\ell})^{\delta_j}. \tag{2.8}$$

Lemma 2.4. *Up to the sign of a block-column permutation, the Gram determinant of $\mathcal{V}_i$ equals* $\det(H_i(t))^{\delta_i}$.

Proof. Order the blocks by increasing s. The (s, s') block has entries

$$\sum_{\ell=1}^{n} w_\ell z_{i,\ell}^{s+s'} h_{i,\ell}^{d_i-s-s'} \prod_{j\neq i} z_{j,\ell}^{\varepsilon_j+\eta_j} h_{j,\ell}^{\delta_j}. \tag{2.9}$$

If $s + s' \le d_i$, this is a polynomial expression in the moment box, and all its coordinate-i degrees exceed b_i. Hence the entire block is zero. If $s + s' = d_i + 1$, the block is exactly $H_i(t)$.

Write $s = b_i + 1 + a$, where $0 \le a < \delta_i$. After reversing the block columns, the column with index a' corresponds to $s' = d_i - b_i - a'$, so

$$s + s' = d_i + 1 + a - a'.$$

Every block above the diagonal is therefore zero, while every diagonal block is $H_i(t)$. The resulting block lower triangular matrix has determinant $\det(H_i(t))^{\delta_i}$, as asserted. □

The next identity proves that H_i is nonsingular as a rational matrix. Its form is chosen to include zero node coordinates.

Lemma 2.5. *For integers $d \ge 2b$ and $b \ge 1$, the following identity holds whenever $t \ne 0$ and $1 + tz \ne 0$:*

$$\frac{z^{d+1}}{1+tz} = \sum_{p=b}^{d} (-1)^{d-p} \frac{z^p}{t^{d+1-p}} + (-1)^{d+1-b} \frac{z^b}{t^{d+1-b}(1+tz)}. \tag{2.10}$$

For each fixed $z \ne 0$, the final term has order $O(t^{-(d-b+2)})$ at infinity. For $z = 0$, every term in (2.10) vanishes identically.

Proof. Multiply (2.10) by $(1+tz)t^{d+1-b}$. Consecutive terms of the finite sum cancel, leaving exactly the numerator on the left. If $z \ne 0$, the additional denominator $1 + tz$ gives one further power of t^{-1} at infinity. If $z = 0$, the factor z^b vanishes because $b \ge 1$, as do all the other numerators. Thus the identity and the remainder statement require no division by a node coordinate. □

Lemma 2.6. *For every i with $\delta_i > 0$, $\det H_i(t)$ is a nonzero element of K.*

Proof. Expand (2.8) at $t_i = \infty$, treating the other parameters as elements of the coefficient field $\mathbb{C}(t_j : j \ne i)$. Apply (2.10) separately to each node. The sum is finite, and its zero-coordinate terms vanish identically, so the remainder estimate applies entrywise. Every term with $b_i < p \le d_i$ has zero coefficient by (2.2) and (2.1). The coefficient with $p = b_i$ is $\beta_i \bigotimes_{j \ne i} G_j(t_j)$. Consequently

$$H_i(t) = (-1)^{d_i - b_i} \beta_i \, t_i^{-(d_i - b_i + 1)} \bigotimes_{j \ne i} G_j(t_j) + O(t_i^{-(d_i - b_i + 2)}). \tag{2.11}$$

This may equivalently be read as a formal Laurent expansion in t_i^{-1}. By Lemma 2.2, its leading coefficient matrix is invertible. The coefficient of $t_i^{-(d_i - b_i + 1)r_i}$ in $\det H_i$ is therefore the nonzero determinant of that matrix. In particular, $\det H_i$ cannot be the zero rational function. Each coordinate is treated separately; no common limiting path for the parameters is required. □

Proof of Theorem 2.1. Choose one finite parameter vector on which every required block is nonsingular. To justify that choice explicitly, put

$$D_i(t_i) = \prod_{\ell=1}^{n} (1 + t_i z_{i,\ell}).$$

For $\delta_i > 0$, the entries of $D_i H_i$ are polynomials in all the parameters. Hence

$$P_i(t) := D_i(t_i)^{r_i} \det H_i(t)$$

is a nonzero polynomial by Lemma 2.6. The product

$$\left(\prod_{i=1}^{k} D_i(t_i) \right) \left(\prod_{i:\, \delta_i > 0} P_i(t) \right)$$

is nonzero. Since $\mathbb{C}$ is infinite, it is nonvanishing at some finite $t \in \mathbb{C}^k$. Fix such a choice. All the vectors (2.5)–(2.6) and the pairing (2.4) are then defined. The Gram matrix of $\mathcal{U}$ is invertible by Lemma 2.2, and each nonempty $\mathcal{V}_i$ has an invertible Gram matrix by Lemmas 2.4 and 2.6. Lemma 2.3 makes their combined Gram matrix block diagonal and invertible.

The combined list thus consists of $Q+\sum_i \delta_i Q/q_i = N$ linearly independent vectors in $\mathbb{C}^n$, proving $n \geq N$. No node was discarded or required to be distinct or to have nonzero coordinates. If a moment formula initially contains zero weights, those terms may first be deleted. □

3 From decompositions to product moments

We now deduce the lower bound in Theorem 1.1 from Theorem 2.1. This reduction begins with an arbitrary decomposition; no affine-support hypothesis is imposed on it.

Proposition 3.1. *Under the hypotheses of Theorem 1.1, every decomposition of $F = \prod_i x_i^{a_i} y_i^{b_i}$ has at least $N(\mathbf{a}, \mathbf{b})$ summands.*

Proof. Fix a decomposition (1.1) of length n and delete any zero summands. Thus every c_ℓ and every pair $(u_{i,\ell}, v_{i,\ell})$ is nonzero. For each i, choose $\gamma_i \in \mathbb{C}$ so that

$$u_{i,\ell} + \gamma_i v_{i,\ell} \neq 0 \qquad (1 \leq \ell \leq n). \tag{3.1}$$

If $v_{i,\ell} = 0$, this condition is automatic; otherwise it excludes one value of γ_i. Only finitely many values are excluded, so the choice exists. Set

$$z_{i,\ell} = \frac{v_{i,\ell}}{u_{i,\ell} + \gamma_i v_{i,\ell}}, \qquad w_\ell = c_\ell \prod_{i=1}^{k} (u_{i,\ell} + \gamma_i v_{i,\ell})^{d_i}. \tag{3.2}$$

All weights are nonzero. Nodes may repeat, and $z_{i,\ell}$ is allowed to be zero.

For $0 \leq e_i \leq d_i$, apply the differential operator

$$\prod_{i=1}^{k} (\partial_{x_i} + \gamma_i \partial_{y_i})^{d_i - e_i} \partial_{y_i}^{e_i}$$

to the decomposition and divide by $\prod_i d_i!$. On its ℓth summand the result is $w_\ell \prod_i z_{i,\ell}^{e_i}$. On the target monomial the result factors over the blocks. The one-variable functionals obtained in this way satisfy

$$m_i(z_i^e) = \begin{cases} \dfrac{(d_i - e)!\, b_i!}{(b_i - e)!\, d_i!} \gamma_i^{b_i - e}, & 0 \leq e \leq b_i, \\ 0, & b_i < e \leq d_i. \end{cases} \tag{3.3}$$

Indeed, when $e \leq b_i$, differentiation by $\partial_{y_i}^e$ first contributes $b_i!/(b_i - e)!$. In the remaining binomial expansion of $(\partial_{x_i} + \gamma_i \partial_{y_i})^{d_i - e}$, only the term with $b_i - e$ derivatives in y_i and a_i derivatives in x_i survives. Its value gives (3.3). For $e > b_i$ the first differentiation already vanishes.

Consequently the product-moment identities hold throughout the entire box $0 \leq e_i \leq d_i$, and

$$\beta_i = m_i(z_i^{b_i}) = \frac{a_i!\, b_i!}{d_i!} = \binom{d_i}{b_i}^{-1} \neq 0.$$

Because $a_i \geq b_i$, we have $d_i \geq 2b_i$. Theorem 2.1 gives

$$n \geq \prod_i (b_i + 1) + \sum_i (d_i - 2b_i) \prod_{j \neq i} (b_j + 1) = N(\mathbf{a}, \mathbf{b}),$$

as required. □

The initial chart (3.1) and the auxiliary parameters in the moment proof serve different purposes. The first makes an arbitrary finite decomposition affine. The second makes the Gram blocks nondegenerate. Both choices are made after fixing the decomposition, and both follow from explicit finite nonvanishing conditions. Neither is a genericity assumption on the decomposition.

4 A finite-character decomposition

We construct a decomposition that attains the proposed upper bound. Let $k \geq 1$, let $a_i \geq b_i \geq 1$ be integers, and set

$$F = \prod_{i=1}^{k} x_i^{a_i} y_i^{b_i}, \qquad d_i = a_i + b_i, \qquad q_i = b_i + 1, \qquad \delta_i = a_i - b_i.$$

Write

$$Q = \prod_{i=1}^{k} q_i, \qquad N = Q + \sum_{i=1}^{k} \delta_i \frac{Q}{q_i}. \tag{4.1}$$

The construction uses a lattice whose nonzero vectors avoid the box of possible exponent differences, followed by character orthogonality on the corresponding finite quotient.

Lemma 4.1. *Let $\delta = (\delta_1, \dots, \delta_k)^T$ and $\mathbf{1} = (1, \dots, 1)^T$. Define*

$$A = \operatorname{diag}(q_1, \dots, q_k) + \delta \mathbf{1}^T, \qquad \Lambda = A\mathbb{Z}^k.$$

Then $\det A = N$, so $[\mathbb{Z}^k : \Lambda] = N$, and

$$\Lambda \cap \prod_{i=1}^{k} [-a_i, b_i] = \{0\}. \tag{4.2}$$

Proof. The lattice is generated by the *columns* of A: its jth generator is $q_j \mathbf{e}_j + \delta$, where $\mathbf{e}_1, \dots, \mathbf{e}_k$ are the standard basis vectors. Expand the determinant multilinearly in these columns. Any term that chooses δ from two columns vanishes. The term choosing no such column is Q, and the term choosing it only from column i is $\delta_i \prod_{j \neq i} q_j$. Hence

$$\det A = Q + \sum_{i=1}^{k} \delta_i \prod_{j \neq i} q_j = N > 0.$$

The index of a nonsingular integer matrix's column lattice is the absolute value of its determinant: integer row and column operations preserve both quantities and reduce the matrix to diagonal form, where the claim is immediate. Thus $[\mathbb{Z}^k : \Lambda] = N$.

Suppose $u = Az$ belongs to the box in (4.2), with $z \in \mathbb{Z}^k$, and put $S = \sum_i z_i$. The column convention gives

$$u_i = q_i z_i + \delta_i S. \tag{4.3}$$

If $S > 0$, then $u_i \leq b_i = q_i - 1$ and $\delta_i \geq 0$ imply $q_i z_i \leq q_i - 1$ for every i. Thus every $z_i \leq 0$, contrary to $S > 0$.

If $S < 0$, write $H = -S \geq 1$. Since $a_i = q_i - 1 + \delta_i$, the lower bound $u_i \geq -a_i$ gives

$$q_i z_i \geq -q_i + 1 + (H - 1)\delta_i \geq -q_i + 1.$$

Integrality now implies $z_i \geq 0$ for every i, contrary to $S < 0$. Finally, if $S = 0$, the upper bounds again imply $z_i \leq 0$ for all i, and their zero sum forces $z = 0$. This proves (4.2), including the cases in which some δ_i vanish. □

Lemma 4.2 (Finite character orthogonality)**.** *Let H be a finite abelian group of order N, and let $\widehat{H} = \mathrm{Hom}(H, \mathbb{C}^\times)$. Then $|\widehat{H}| = N$ and, for every $h \in H$,*

$$\sum_{\chi \in \widehat{H}} \chi(h) = \begin{cases} N, & h = 0, \\ 0, & h \neq 0. \end{cases} \tag{4.4}$$

Proof. Write H as a product of cyclic groups $\mathbb{Z}/m_1\mathbb{Z} \times \cdots \times \mathbb{Z}/m_s\mathbb{Z}$. A character is determined by independently choosing an m_jth root of unity as the image of the jth cyclic generator. There are therefore $\prod_j m_j = N$ characters. If h has coordinates h_j, the sum in (4.4) factors as

$$\prod_{j=1}^{s} \left(\sum_{r=0}^{m_j - 1} \exp\!\left(\frac{2\pi \mathrm{i}\, r h_j}{m_j} \right) \right).$$

Each geometric sum equals m_j when $h_j = 0$ in $\mathbb{Z}/m_j\mathbb{Z}$ and equals zero otherwise, proving the formula. □

Proposition 4.3. *The polynomial F admits a decomposition into N nonzero weighted products of powers of linear forms, with degree d_i in the ith block. In particular, its fixed-block rank is at most N.*

Proof. Use the column lattice Λ of Lemma 4.1, and put

$$H = \mathbb{Z}^k/\Lambda, \qquad G = \widehat{H}, \qquad \theta_i(\chi) = \chi(\mathbf{e}_i + \Lambda) \quad (\chi \in G).$$

Each $\theta_i(\chi)$ is a nonzero root of unity. We claim the exact identity

$$\boxed{F = \frac{1}{N \prod_{i=1}^{k} \binom{d_i}{a_i}} \sum_{\chi \in G} \left(\prod_{i=1}^{k} \theta_i(\chi)^{-a_i} \right) \prod_{i=1}^{k} \left(\theta_i(\chi) x_i + y_i \right)^{d_i}.} \tag{4.5}$$

Indeed, let $m = (m_1, \ldots, m_k)$ satisfy $0 \le m_i \le d_i$. The coefficient of $\prod_i x_i^{m_i} y_i^{d_i - m_i}$ on the right of (4.5) is

$$\frac{\prod_i \binom{d_i}{m_i}}{N \prod_i \binom{d_i}{a_i}} \sum_{\chi \in G} \chi(m - a + \Lambda), \qquad a = (a_1, \ldots, a_k). \tag{4.6}$$

Since $d_i - a_i = b_i$, the vector $m - a$ lies in $\prod_i [-a_i, b_i]$. Lemma 4.1 says that it belongs to Λ precisely when $m = a$. Character orthogonality therefore makes (4.6) zero for $m \neq a$, while for $m = a$ it equals one. This verifies every coefficient of (4.5).

There are $|G| = N$ summands, and every weight in the displayed formula is nonzero. The normalized support tuples are also distinct: the images of the standard basis generate H, so two characters with the same θ_i for every i coincide. Moreover, the product $\prod_i (\theta_i x_i + y_i)^{d_i}$ has all-y coefficient one and coefficient $d_i \theta_i$ at $x_i y_i^{d_i - 1} \prod_{j \neq i} y_j^{d_j}$; thus distinct tuples give nonproportional products. The identity is consequently an N-term fixed-block decomposition over $\mathbb{C}$, and proves the upper bound. □

Proof of Theorem 1.1. Proposition 3.1 bounds every decomposition below by $N(\mathbf{a}, \mathbf{b})$. Proposition 4.3 supplies a decomposition with exactly this many summands. □

5 Specializations and scope

For one block, (1.2) is $\mathrm{R}(x^a y^b) = a + 1$ when $a \geq b \geq 1$. For two blocks it becomes

$$\begin{aligned} N &= (b_1 + 1)(b_2 + 1) + (a_1 - b_1)(b_2 + 1) + (a_2 - b_2)(b_1 + 1) \\ &= (a_1 + 1)(a_2 + 1) - (a_1 - b_1)(a_2 - b_2). \end{aligned}$$

These are the one-block and two-block overlaps described above. When all $a_i = b_i$, the additional moment lists are absent and the formula reduces to $N = \prod_i (b_i + 1)$; the lattice matrix in the upper construction is then diagonal.

Some immediate examples are recorded in Table 1. They are consequences of the quantified theorem, not an interpolation from finitely many rank computations.

Fixed-block tensor or polynomial	Complex partially symmetric rank
$x^a y^b$, $a \geq b \geq 1$	$a + 1$
$\prod_i x_i^{b_i} y_i^{b_i}$	$\prod_i (b_i + 1)$
$W_3^{\otimes k}$, $k \geq 2$	$2^{k-1}(k + 2)$
$W_3 \otimes W_3 \otimes W_3$	20
$(x^3 y^2)^{\otimes 3}$	54

Table 1: Special cases of Theorem 1.1. Each tensor power keeps its displayed factors as symmetric blocks.

For the last example $q_i = 3$ and $\delta_i = 1$, so $N = 3^3 + 3 \cdot 3^2 = 54$. Its lattice matrix has diagonal entries 4 and off-diagonal entries 1, and (4.5) provides the 54 summands. For the cubic W triple the corresponding matrix has diagonal entries 3 and off-diagonal entries 1, and the number of characters is 20.

The hypothesis $b_i \geq 1$ is essential to the theorem as stated. It is used when the Laurent remainder is evaluated at a zero node. The formula would already be false for the single pure power x^a with $b = 0$: its rank is 1, whereas the right-hand side would be $a + 1$ for $a \geq 1$. Such factors are excluded from the statement.

All Gram forms in the proof are symmetric bilinear forms over $\mathbb{C}$. No positive weights or Hermitian positivity are assumed. Invertibility of every diagonal Gram block, established before dimensions are added, is what prevents isotropic or radical subspaces from invalidating the count. The Laurent expansions certify nonzero rational functions; the final argument evaluates those functions at one finite point. No limiting decomposition enters the proof.

A Exact finite checks

The proof above establishes the theorem for all its stated parameters without a computer calculation. Two finite sets of exact checks provide additional arithmetic checks on its formulas. Their scope is described here to distinguish them from the universal proof.

For the W-state specialization, rational calculations in degrees 2 through 9 check the univariate moment and Gram identities. A calculation over $\mathbb{Q}(\zeta_{10})$ verifies all coefficients of a 20-term cubic triple decomposition and an invertible 20×20 Gram matrix at an asymmetric finite parameter vector. Rejection checks change a coefficient, choose a pole, and impose an incorrect nondegeneracy assertion at a symmetric chart. The last check rejects that chart, while leaving the validity of the decomposition intact.

For the general binary-monomial construction, integer, rational, and cyclotomic arithmetic verifies all coefficients in eight explicit character decompositions, including the 54-term example, unequal degrees, a balanced factor, four blocks, and the one-block, two-block, and cubic-W overlaps. Three rational cubatures check their full Gram matrices directly, with zero and repeated nodes included. Separate identities check the Laurent remainder at positive, negative, and zero rational nodes. Eight rejection checks alter, respectively, the rank count, lattice orientation, character list, character value, target phase, cubature weight, Gram-vector list, and remainder sign.

These finite computations use no floating-point rank tests. They cannot replace the finite-chart existence proof, the all-parameter lattice argument, or the reduction of an arbitrary decomposition to moments. The mathematical conclusions do not rely on an assumption that the checked examples exhaust the possible supports.

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TBD

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